Local opportunity does not ensure end-to-end gain

Observed case	Favorable local measurement		Realized outcome
M4 · HumanEval30 Preregistered, fixed k = 2	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + profitable** 1.387× · 90/90 exact
M4 · representative 24Q Frozen representative study	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + negative** 0.842× · 72/72 exact
RTX 5070 Ti Laptop CUDA batched verification	$V_2^{\text{micro}} = 0.983–1.009$ Near one target-decode step	→	□ **Correctness blocked** No admissible speedup
358H · NPU draft Qualified Phase-3 path	48.08 vs 30.81 tokens/s NPU vs CPU · isolated, 8,192 tokens	→	● **Exact + negative** 0.352× · 90/90 exact

Categorical cases, not a hardware ranking. Filled = qualified; open = blocked. Details and CIs: Figures 2–4.

Micro cost ≠ complete target work; isolated throughput ≠ stateful cost. Speedups cover post-prefill decode.